

Collect a shared trajectory group



Same shopping task x + initial state S
 N environment copies; one rollout policy



Old policy

 $\pi_{\theta_{\text{old}}}$

action

Environment



state + reward

Trajectory

Observed state : action

Return

 τ_1 S $t = 1$

2nd

 P

back

 S

1st

 $t = 3$ 

1

 τ_2 S $t = 1$

next

 Q

...



end

-

 $t = 3$ 

0

State key $S \rightarrow$ all matching occurrences (τ_1, t_1) (τ_1, t_3) (τ_2, t_1) 

Same state, different times / trajectories.

Offline hash grouping reuses collected actions.



Clipped action objective + reference KL

 p : current / old action probability

Episode & anchor-state comparisons



Episode relative advantage

$$E_i = \sum_t r_t^{(i)}, \quad A_i^E = \frac{E_i - \mu_E}{F_E}$$

Compare total returns across trajectories



Step relative advantage

$$Q_{i,t} = \sum_{k=t}^{T_i} \gamma^{k-t} r_k^{(i)}$$

Future return, not just immediate reward

$$A_{i,t}^S = \frac{Q_{i,t} - \mu_{\mathcal{J}(S)}}{F_{\mathcal{J}(S)}}$$

Center within the matching-state group

Normalization F: group std or 1

Fuse credit & update



Global + local

$$A_{i,t} = A_i^E + \omega A_{i,t}^S$$

 ω balances the two signals

Different steps can get different credit.



Trainable actor

 π_{θ}

Clipped update



Reference policy

 π_{ref}

KL anchor



Action-specific credit

No extra per-state rollout or critic

$$\hat{J} = \frac{1}{NT} \sum_{i,t} \min(\rho_{i,t} A_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_{i,t}) - \beta D_{\text{KL}}(\pi_{\theta} \| \pi_{\text{ref}})$$

KL: keep the actor close to the reference